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An outcome counting based solution to the problem of probability in the many worlds interpretation

Outcome counting and the MWI

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Abstract The chief conceptual difficulty with the many worlds interpretation of quantum mechanics is the so-called “problem of probability” that accompanies the assumption of the Born rule. What does it mean for two Everettian branches with unequal Born measure to be equally “real,” yet have unequal probability? Advocates of Deutsch-Wallace decision theory claim to have solved this problem by deriving the Born rule from the “non-probabilistic” axioms of quantum theory and classical decision theory. In this paper, a derivation of quantum probabilities is proposed that is distinct from decision theory in two fundamental respects. 1) The circular reasoning that is (arguably) inherent to the decision theoretic program is avoided in the proposed model because *none* of the axioms of quantum theory – probabilistic or not – are included in the list of starting assumptions. 2) The proposed model replaces the Born rule at the fine-grained level with the most natural probability rule one would hope to associate with the MWI, outcome counting, which is the notion that each branch of an Everettian tree is not only equally real, but also *equally likely*. Starting from a relational reading of the MWI, the proposed scheme assumes a classical mechanical representation of the observer, and it is argued that outcome counting gives rise naturally to an *action principle* that may provide a basis for the derivation of quantum probabilities and the ultimate recovery of the Born rule as a coarse-grained approximation. The resulting framework outlines a solution to the problem of probability of the MWI that is more natural and satisfying than that proposed by decision theory.

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1 Introduction

In the fifty years since the publication by Hugh Everett of the relative-state formulation [1] of quantum mechanics – more commonly known as the many-worlds interpretation (MWI) – his insights have been gradually gaining acceptance among physicists and philosophers of science. Nevertheless, thorny philosophical problems remain, even among its adherents. Perhaps the most difficult problem has been to justify the interpretation of the Born rule measure as a probability.

Recently, much attention has been focused on a program originated by David Deutsch [2] and advanced by Wallace [3], Saunders [4], and others to use decision theory to solve the Everettian problem of probability. The goal of Deutsch-Wallace decision theory is to derive the Born rule from the non-probabilistic axioms of quantum theory – namely, the usual Hilbert space structure of a quantum system – and the non-probabilistic axioms of classical decision theory. Critics of this program maintain not only that Deutsch-Wallace decision theory is guilty of circular reasoning [5], but also that it fails to account for the fact that on a simple count of the worlds involved, the vast majority of observers are so-called “atypical” in the sense that their experience *contradicts* theoretical predictions [6].

The purpose of this paper is to propose an alternative strategy for the derivation of quantum probabilities that avoids these critiques. First, the proposed model eliminates suspicion of circular reasoning by eliminating *all* of the axioms of quantum mechanics – probabilistic or not – from the list of starting assumptions, in favor of classical assumptions. Second, the proposed model implements the most obvious algorithm for generating probabilities in the MWI, outcome counting, which is the notion that each branch of an Everettian tree is equally likely. By this probability rule, the vast majority of observers are “typical” in the sense that their experience *confirms* theoretical predictions [6].

Beginning with a relational reading of the many worlds interpretation that emphasizes the role of the observer [7], a model is proposed that assumes 1) a classical description of the state of the observer, and 2) a probability law based upon outcome counting. Standard quantum mechanical axioms such as a Hilbert space or the Schrödinger equation are notably absent from the starting assumptions. The primary result of this paper is that the proposed model provides a natural framework for *derivation* of one of the most powerful statements in physics: an *action principle*. It is then speculated that there exists a suitable definition of the action from which will follow the equations of quantum field theory¹. In this way, the core conceptual difficul-

¹ This is meant in the same sense that the Einstein equations follow from the Einstein-Hilbert action[9].

ties surrounding the probability interpretation of the MWI are solved in a manner that is much more straightforward and satisfying than that provided by decision theory.

2 A classical representation of matter

An individual Everettian world is given a classical description as a classical 4-dimensional spacetime – that is, a 4-manifold imbued with a metric g_{ij} – comparable to a history in the consistent histories program [8]. The metric g_{ij} is assumed to be analytic, and there are no additional matter fields that exist on top of spacetime; rather, “all is geometry.” The topology is not restricted to the simply connected case; multiply connected manifolds are allowed.

The smallest and most elementary “chunk” of matter that can be imagined is an arbitrarily small, simply connected 4-ball with point p at its center. Specification of the metric g_{ij} throughout the 4-ball allows the calculation of the metric and power series at p . Conversely, provided with the metric and its complete power series at p , the method of analytic continuation allows the calculation of g_{ij} throughout the entire 4-ball. Indeed, analytic continuation can be used to calculate the metric throughout an entire 4-manifold, provided 1) the metric and power series at p , and 2) the topology of the 4-manifold.

In accordance with Everett’s original paper (and as emphasized by Rovelli’s relational reading [7] of the MWI), the role of the observer can be played by any physical system. In the present construction, any (classical) physical object – including an observer in the traditional sense, such as a computer or a human brain – can be constructed using the 4-balls (with metric) as building blocks. These elementary building blocks will serve in the present construction as the most elementary example of an “observer” that can be considered. The set of all possible 4-ball plus metric defines the set of all possible *elementary observers*. One elementary observer O corresponds to a point in an infinite dimensional state space, with each dimension being one component of the metric or its power series.

Consider the set T of “all possible 4-topologies.” Given any single elementary observer O , each distinct topology results (through analytic continuation) in a distinct spacetime. The set of all such solutions, M_o , can aptly be described as the set of Everettian worlds (spacetimes) *relative to* the system O . Nothing other than O is required to specify M_o in full. Recall Everett’s original paper, in which he states that “one can arbitrarily choose a state for one subsystem, and be led to the relative state for the remainder” [1]. In the proposed model, this is interpreted to mean: “one can arbitrarily choose a state O for one subsystem, and be led to the relative state M_o for the remainder.”

3 Time-dependent evolution of the observer

Classically, the evolution in time of a physical object is described by its propagation along a geodesic, as calculated by the parallel transport law, i.e. according to the geodesic equations [9]:

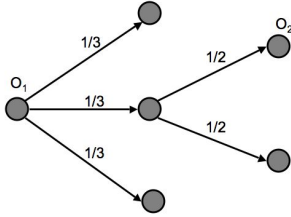


Fig. 1 Outcome counting in an Everettian tree. If a tree branches into N directions, then the probability associated with each branch is equal to $1/N$. The probability rule is transitive, so that the probability of evolution from O_1 to O_2 is $\frac{1}{3} \frac{1}{2} = \frac{1}{6}$

$$\frac{d^2 x^k}{dt^2} + \Gamma_{ij}^k \frac{dx^i}{dt} \frac{dx^j}{dt} = 0 \quad (1)$$

In the proposed scheme, these same equations will be used to define the evolution of the observer O in time within any given spacetime. That is, for a given spacetime, the geodesic equations will yield: $O(\tau) = g_{ij}(\tau)$ where τ is the proper time of the observer.²

For each spacetime (for each element of M_o), there will be a distinct evolution $g_{ij}(\tau)$.³ For any particular $O = g_{ij}(0)$, there is a unique set of all possible evolutions $g_{ij}(\tau)$. If these evolutions are drawn on a graph with g_{ij} as the ordinate and τ as the abscissa, the result (in this construction) is the Everettian tree. For any point O in state space, there is a unique Everettian tree rooted at that point.

4 Probability

The common view of the MWI is that all branches are, in some sense, equally “real.” According to this ontology, one might expect that each branch should be ascribed an equal likelihood. This notion (ascribing equivalent likelihood to each branch of an Everettian tree) has gone by several names in the literature (including egalitarianism[10] or the “alternate projection postulate”[11]), but, following Weissman[12], will be referred to in this paper as “outcome counting” (see Figure 1).

Many authors have noted that outcome counting, rather than the Born rule, is the most natural probability rule that one would hope to implement

² It should be remarked that a complete definition of the elementary observer O includes a unit vector pointing in (defining) the forward time direction in addition to specification of the metric and power series. We are interested in $g_{ij}(\tau)$ only for $\tau \geq 0$.

³ Typically, any two evolutions $g'_{ij}(\tau)$ and $g''_{ij}(\tau)$ may be identical up to some time τ_b , where b indicates “branch point,” but nonidentical after; that is, for example, that $g'_{ij}(\tau) = g''_{ij}(\tau)$ for $0 \leq \tau \leq \tau_b$, but $g'_{ij}(\tau) \neq g''_{ij}(\tau)$ for $\tau > \tau_b$.

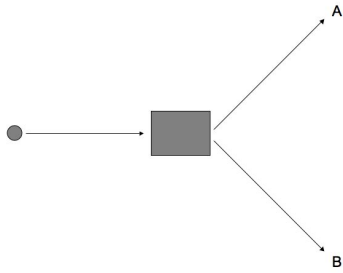


Fig. 2 Multiplicity of solutions in classical mechanics. “Outcome counting” is the notion that an observer represented by the circle would expect 50:50 odds of evolution along trajectory A or B.

in the MWI. As stated by Graham, “it is extremely difficult to see what significance such a measure [the Born rule measure] can have when its implications are completely contradicted by a simple count of the worlds involved, worlds that Everett’s own work assures us must all be on the same footing” [13]. Referring to this difficulty as the “quantitative problem,” Greaves writes: “The quantitative problem is perhaps unproblematic in the case of equally weighted superpositions, but has been considered fatal in the case of unequally weighted superpositions. . . . [I]t seems that all Everett can say is that each outcome occurs in exactly one branch, which, we might think, will yield equal probabilities if any at all” [10].

The proposed scheme adopts outcome counting as a postulate. This probability rule is illustrated by the Everettian tree in Figure 1. It is assumed that this rule is transitive: thus, the probability of evolution from O_1 to O_2 is $\frac{1}{3} \frac{1}{2} = \frac{1}{6}$.

5 The classical foundation for a probabilistic theory

A strong motivation for basing a probabilistic theory on purely classical principles comes from the demonstration of “multiplicity” within classical mechanics. Thorne and colleagues [14] pointed out that under certain situations it is possible, in classical general relativity, to describe a system in a single, known, definite classical state, such that there is *more than one solution to the time dependent evolution of the system consistent with the laws of motion*. The number of distinct solutions was termed the “multiplicity” of the problem, and might be finite or infinite. The only feature that made their spacetimes unusual was that they allowed closed timelike curves (CTCs) – i.e., wormholes or handles in the fabric of spacetime. This feature (allowing multiply connected spacetime) is also allowed in the model proposed in this paper.

Figure 2 illustrates a typical example of multiplicity. The system in question is a billiard ball (the circle) that interacts with another system (the box) containing closed timelike curves. In this example, there are two solutions to

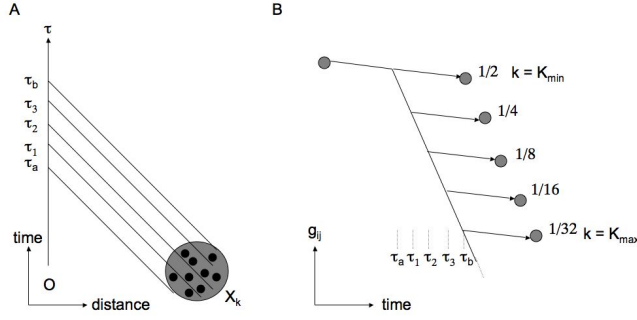


Fig. 3 A. O cannot make internal recordings of the state of X_k until X_k is in the past light cone of O . In this example, k is the total number of black dots (microscopic CTCs) in the interior of X . B. The Everettian tree corresponding to A. Shaded circles indicate observer states. Each branch to the right corresponds to a different final internal recording value of k . Probabilities of arriving at certain states are indicated, as calculated by the outcome counting rule.

the trajectory of the billiard ball, represented by the two curves A and B. These trajectories overlap before the interaction between circle and box, but differ after the interaction. In this case, there is multiplicity = 2.

If the billiard ball plays the role of observer in the proposed scheme, then the corresponding Everettian tree should look like the trajectory diagram of Figure 2. The principle of outcome counting would then predict that this observer, prior to reaching the box, should expect a 50 percent chance of movement along trajectory A, and likewise 50 percent for B.

6 An action principle

Consider an observer O that is spacelike separated from a system X_k under observation, and that X_k is in one of a multiplicity K of states $k \in [K_{min}, K_{max}]$ where k is, for example, the number of microscopic CTCs found in the interior of the boundary of X_k (or perhaps some other topological invariant). We wish to know what the Everettian tree rooted at O looks like as it observes X_k .

Relativity tells us that O cannot “observe” X_k – i.e. cannot create an internal recording of the value of k – until X_k is in the past light cone of O (see Figure 3A). X_k is an extended (small but finite; not point-like) object and therefore enters the past light cone over an extended time interval $[\tau_a, \tau_b]$ – not all at once. Over this time interval, O can be imagined to “integrate over” the entire volume of X_k to arrive at the total value k . At any given time, k_τ is the number of CTCs that have so far fallen within the past light cone. In other words, k_τ is like a “running total” that gets incremented bit by bit until it reaches some final value k . Each forward iteration in time yields one of two possibilities (resulting in a series of bifurcations in the Everettian tree of Figure 3B): 1) The surface boundary of X has been engulfed in the past light cone of O , yielding a final value for k (represented by the rightward

branches in Figure 3B); or 2) the surface of X has *not* been engulfed by the past light cone, so that the final value of k is not yet known but must take some value in the interval $[k_\tau, K_{max}]$ (represented by the downward-sloping branches in Figure 3B).

The resulting Everettian tree is that depicted in Figure 3B. The next step is to apply outcome counting to calculate the probability of arriving at a particular state O_k . The remarkable result is that lower values of k are more likely than higher values. That is to say that k has been *minimized*.

7 Application of the action principle

In the above fashion it is proposed that various action principles may be constructed using various definitions of the variable k . It is well known that many important results in physics can be derived via an action principle, well established examples including Newton's laws[15] (via Hamilton's principle of least action) and the Einstein equations[9]. The general way to define an action principle is to define an action $S = \int L$ (where L is a lagrangian or lagrange density) and to postulate that S takes its minimum value (or more generally, extreme value). It should be noted that the notion that action is minimized (or stationary) is typically introduced as a *freestanding axiom*. In the preceeding discussion, k plays the role of the action S . Its minimization is a *consequence* of the probability rule (outcome counting) and need not be assumed independently.

The definition of an action principle hinges on identification of the appropriate mathematical expression for L . Although it is not demonstrated in this paper, it is speculated that there exists a suitable definition of L such that 1) its minimization is a consequence of the probability rule, and 2) its minimization yields quantum mechanics (or quantum field theory).

8 Summary

Unlike previous attempts to solve the problem of probability in the MWI, the model proposed in this paper avoids circular reasoning because it is founded upon purely classical axioms that bear no resemblance to *any* of the axioms of quantum theory. This should be evident by a brief review of the proposed model's assumptions:

1. A "world" is a spacetime, i.e. a manifold imbued with a metric g_{ij} , and a physical object is a region within a spacetime. There are no additional fields. The metric g_{ij} is analytic. Manifolds are 4-dimensional and may or may not be time oriented. The manifold is continuous and not necessarily simply connected: multiple connectedness is allowed.
2. The time-dependent evolution of a physical object within a spacetime is governed by the parallel transport law. "Time" indicates proper time of the object. (As already mentioned, spacetimes are not necessarily assigned a prespecified time-orientation.)
3. Given a multiplicity of possible evolutions, probabilities are calculated by outcome counting.

The proposed model does *not* list the following as assumptions: the Schrödinger equation; the Born rule; the projection postulate; that states must be elements of Hilbert space; that state evolution is unitarity, nonunitarity, linear, or nonlinear; that branch fusion is or is not allowed.

In principle, any conceivable experiment can be phrased as the question: given an observer in state O_1 , what is the probability of evolution into state O_2 ? The question is answered using the following procedure:

1. Define an observer O_1 .
2. Calculate the collection M_{O_1} of all possible worlds relative to O_1 .
3. Calculate all possible geodesics beginning at O_1 by the parallel transport law. The probability of evolution into state O_2 is nonzero if and only if at least one geodesic leads from O_1 to O_2 .
4. Draw the Everettian tree relative to O_1 .
5. Apply outcome counting to the tree.

Additional future work is needed to generalize this model from the “elementary” observer O to any generic physical system (i.e., an observer in the classical sense).

It is argued that the proposed model is sufficient to for the derivation of a (generic) action principle, and speculated that there exists a specific action principle (i.e., a specific choice of the lagrangian) from which quantum mechanics (or quantum field theory) will follow as being valid in the appropriate limit. The realization of such a scheme does not seem so farfetched – yet would be a remarkable result given the parsimonious quality of the model’s starting assumptions listed above. And from a philosophical perspective, the proposed model solves the problem of probability in the MWI in a much more satisfactory manner than alternate attempts such as decision theory. These philosophical considerations in and of themselves should provide sufficient motivation for attempts at further development of this model.

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